University of Essex

School of Mathematics, Statistics
and Actuarial Science

MA981 DISSERTATION

Customer churn prediction in telecommunication: Evaluating cross-industry model performance using machine learning and deep learning

RIZWAN
2201583

Supervisor: **Dr Zoe Bartlett**

August 25, 2023
Colchester

Contents

1 Abstract **6**

2 Introduction **7**

3 Literature Review **10**

4 Data set **13**

5 Performance Evaluation **15**

6 Data Preprocessing **17**

6.1 Handling missing data 18

6.2 Data transformation 18

6.3 Feature Selection 18

6.4 Handling Imbalance Data 19

6.5 Exploratory data analysis (EDA) 21

7 Methodology **32**

7.1 Training and Testing Sets for ML Models 33

7.2 Machine Learning For Churn Prediction 33

7.2.1 Support Vector Machine (SVM) 34

7.2.2 Decision Tree 35

7.2.3 Naive Bayes 36

7.2.4 Random Forest Classifier 36

7.2.5 K-Nearest Neighbors (K-NN) 37

7.2.6 Logistic Regression 37

7.2.7 Gradient Boosting 38

7.2.8 Artificial Neural Network (ANN) 38

7.2.9 Convolutional Neural Network (CNN) 39

8 Results and Findings **40**

8.1 Telecommunication Customer Dataset 40

8.2 Bank Customer Dataset 44

9 Conclusions **48**

10 Recommendation **51**

11 Future Work **53**

Bibliography **57**

List of Figures

2.1 Relationship between machine learning and deep learning 9

6.1 Feature selection through Elastic net regression 19
6.2 comparision of dataset before and after balancing 20
6.3 churn rates with respect to gender, senior citizen, partner, and dependents 21
6.4 churn rates with respect to internet service providers 22
6.5 customers who are using fiber optic internet service 23
6.6 customers who are using DSL internet service 24
6.7 churn rates with respect to contract type 26
6.8 churn rates with respect to payment method 27
6.9 churn rates with respect to tenure 28
6.10 churn rates with respect to monthly charges 29
6.11 Top 10 cities which have highest churn rates 30
6.12 customer churn reasons 31

7.1 Machine learning model cycle 32

8.1 churn rates in telecommunication customer dataset 41
8.2 Best machine learning (with all features) and deep learning models performance 44
8.3 churn rates in bank customer dataset 45
8.4 Best machine learning and deep learning models performance 46

List of Tables

4.1 Comparison of Characteristics for Telco Customer and Bank Customer Datasets . . . 14

5.1 Structure of confusion matrix . . . 15

6.1 Churn Rates with respect to Senior Citizen, Partner, and Dependents . . . 22
6.2 Churn Rates with respect to different internet service providers with their services . . . 25
6.3 Churn rate with respect to contract type . . . 27
6.4 Churn rate with respect to payment method . . . 28

8.1 Machine learning and deep learning model performance (Telecom) . . . 43
8.2 Machine learning and deep learning model performance (bank) . . . 47

CHAPTER 1

Abstract

Every company depends entirely on its consumers for revenue and the telecom industry is one of several that are suffering significant losses as a result of customers leaving their services. We can develop machine learning models to assist these businesses in predicting consumers before they churn, allowing them to take better action, and by analyzing their data we can additionally discover the factors that really cause customers to end their relationship with the company and switch to another company. In this thesis, seven machine-learning models which include support vector machine (SVM), decision tree, naive bayes, random forest, K-nearest neighbors (K-NN), logistic regression, gradient boosting, and two deep-learning models which include artificial neural network (ANN), and convolutional neural network (CNN) were applied and had made two of each model one with all features and one with selected features for feature selection elastic net regression was also applied. 10-fold cross-validation was also applied. All of these models were applied to the banking dataset as well to check the transferability of the models. Moreover, both datasets were imbalanced due to dataset imbalance issues smote an oversampling technique was used. The findings indicate that in the telecom dataset, machine learning models with all features outperform models with selected features, and in both datasets, the deep learning models, random forest, and gradient boosting outperform other models. Companies may take more effective action and avoid customer churn if they address a number of important issues that have been discovered in this thesis.

CHAPTER 2

Introduction

In the modern world, we feel extremely close to one another. Whether in our homeland or the most distant corners of the world, all it takes is a few taps on our phone or computer to get in touch with someone. Additionally, from a professional and personal point of view, communication plays an important role in sharing knowledge and ideas. The telecommunications sector has made all of these operations possible. The competition among telecommunications firms is growing, and they employ a variety of tactics to retain and attract customers. Client retention is the industry's biggest concern, thus companies aim to decrease client churn by improving customer satisfaction (Almuqren et al., 2021). Customer churn refers to when a customer decides to end their relationship or subscription with a telecommunications service provider or switch to a different service provider. This may happen for a number of reasons, which include unsatisfactory service, word-of-mouth recommendations, unreasonable costs compared to other service providers, and high prices (Bhattacharyya et al., 2022).

By placing a higher priority on client loyalty, telecommunications firms are turning their attention to retaining the customers they already have instead of finding new customers by doing this they want to establish long-lasting relationships with customers for sustainable growth (Kim et al., 2004). Due to the considerable impact that customer churn has on a company's revenue, companies in the telecom sector are actively looking for a way to predict the CC. To reduce customer churn, it is necessary to identify the elements that contribute significantly to it (Ahmad et al., 2019).

To solve any problem it is necessary to analyze the problem and the data which we have. Before it was not as difficult to manually analyze the data because at that time we did not have that much amount of data it was also time-consuming work. However, since telecommunication companies now generate huge amounts of data daily, including call records, complaints, monthly or weekly packages, call minutes, and many other things, it is almost impossible to manually analyze that data. Machine learning and AI help us to analyze big data by seeing the relationship between features using different methods and also to find some patterns ultimately it saves time. Secondly, because of a large amount of data, AI and machine learning models are more effective than people in identifying hidden trends and forecasting customer churns. Before machine learning and AI actions were taken very late because of the time-consuming process customer behavior can change very frequently over time for instance when companies decide on the final strategy or to take any actions at that time customer data experienced some changes they again need to change their actions before they make them happened but now with the help of AI and machine learning companies can take real-time decisions which helps companies to understand the patterns on time so before churning they can take important actions and make strategies to prevent customer churn. Hence nowadays these organizations need to use AI and Machine learning which helps them to identify customer churn and make new strategies to make their company profitable and make themselves able for sustainable future growth.

In the market there are many models for customer churn (CC) prediction used by telecommunication industries there are some machine learning models that can be used for CC prediction are logistic regression, decision tree, random forest, Gradient boosting, support vector machines (SVM), XGBoost, naive bayes. One thing to keep in mind is the quantity of the data because machine learning algorithms do not perform well in a large amount of data for that we need to use deep learning. Deep learning models exhibit superior performance in numerous scenarios compared to shallow machine learning models and conventional data analysis techniques. The relationship between machine learning and deep learning is shown in figure 2.1 (Janiesch et al., 2021). For the (CC) Prediction we can use Recurrent Neural Networks (RNN), and Convolutional Neural Networks (CNN).

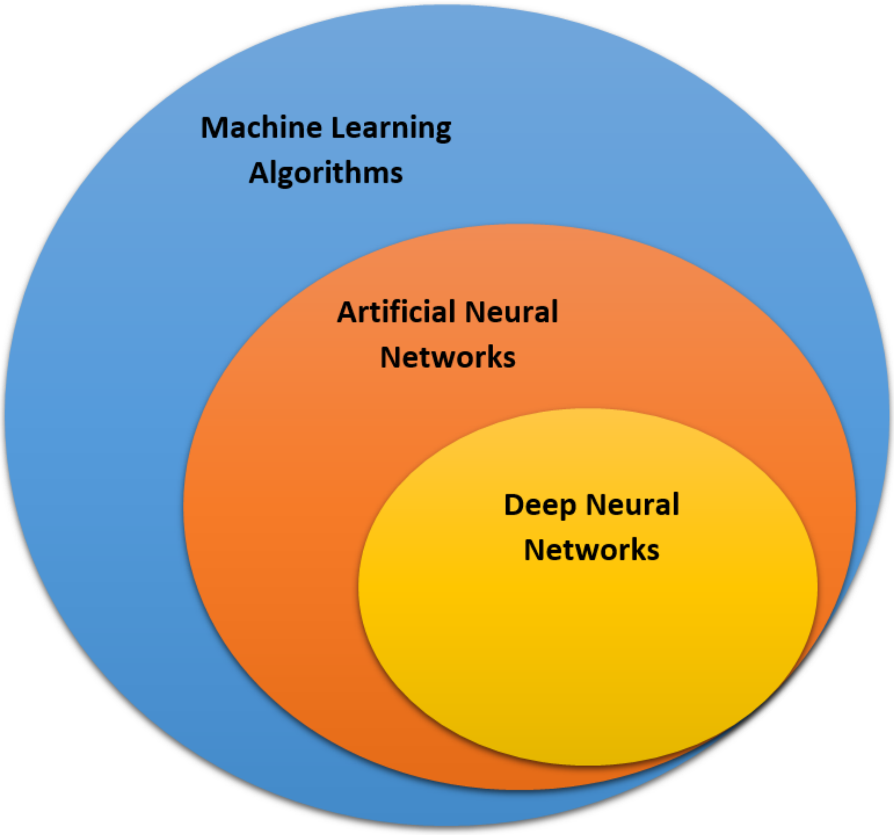

Figure 2.1: Relationship between machine learning and deep learning

This thesis aims to check the transferability of the customer churn predictive models between two different industries which are telecommunication and banking customer loss is the biggest challenge for every industry. Being able to accurately predict which customers leave in the future can help them take better actions in the future to retain their customers by making good strategies and smoothly running their business. Initially, the predictive model for the telecommunication industry was applied, and then the same model in the banking industry to check whether it would perform and give the same performance. Analyzing the performance of the customer churn predictive model across two different industries is very crucial while the customer bases of the banking and telecommunications sectors may differ from one another, there may be some common factors in both that contribute most to customer churn finding these similarities and differences will help to identify the power of the using the same model in a different industry. Additionally, it was also investigated which features play an important role in customer churn in the telecom industry.

CHAPTER 3

Literature Review

Yabas et al. (2012) used the dataset containing 100,000 rows and 15000 features data was a bit noisy and had some missing values preprocessing was done after that author used machine learning and data mining tools for customer churn prediction, the author constructed models using decision trees, random forest, decision trees with bagging and boosting techniques. The random forest model consisting of 50 trees gives the best result with an accuracy of 0.6533 the results indicate that this model performs the best among all achieving the highest accuracy. Vafeiadis et al. (2015) Uses artificial neural networks (ANNs), support vector machines (SVMs), decision trees (DTs), naive bayes classifier, and logistic regression among all methods the two best methods were the back-propagation network with 15 hidden layers and decision tree classifier both methods got 94% accuracy and 77% F-measures. The support vector machine (SVM) with RBF and polynomial kernel function got almost 93% accuracy and 73% F-measure. On the other hand, naive bayes and logistic regression got a lower accuracy of approximately 86% the F-measure of naive bayes was 53% and the F-measure of logistic regression was 14%. All of the model's accuracy was increased because of boosting technique but The boosted SVM (SVM-POLY with Ada boost) got the highest accuracy amongst all other models it got 97% accuracy with 84% F-measure. Jain et al. (2020) used orange telecom companyâs dataset which is an american-based company the dataset contains 3333 rows with 20 features author used the logistic regression model and logit boost both models performed well. The results of both models were almost the same Logistic regression got 85.2385% accuracy whereas the logit boost got 85.1785% accuracy.

Lalwani et al. (2022) Used various machine learning models to compare which model performs well to predict customer churns (CC) models include logistic regression or logit regression, extra tree classifier, decision trees, random forest classifier, naive bayes classifier, CatBoost, Ada â boost, XGBoost and support vector machine(SVM). Three models performed well among all. The decision tree model got an accuracy of 80.14% with 78.81% precision, 80.1% recall, and 78.89% F-measure. The logistic regression model got an accuracy of 80.45% with 79.11% precision, 80.23% recall, and 78.89% F-measure. XGBoost got the highest accuracy of 84%. Huang et al. (2012) Used the dataset containing 827,124 records out of which 13,562 are churners and 400,000 are not going to churn they also used new features for CC prediction which are demographic profiles, information of grants, customer account information, historical information of bills and payments, service orders, henley segments, telephone line information, complaint Information, call details, and incoming calls details they used some machine learning models include Logistic regression, support vector machines (SVM), decision trees, naÃ¯ve bayes, linear classifier, artificial neural networks, and data mining by evolutionary learning they conducted some experiments with features and found that new features perform well compare to existing features compare to other models DMEL was giving very bad performance they also found the models for specific problems like decision trees can identify the churn reasons that why a customer will leave but cannot be able to give the probabilities for that logistic regression, multi-layer perceptron (MLP) and naÃ¯ve bayes models can use where as SVM and LC models can only tell us whether the customer will leave or not. Idris et al. (2012) Used two different datasets one was given by orange telecom company and the other one was cell2cell which is publicly available they applied KNN, random forest, and GP with AdaBoost to predict the customer churns on the orange dataset KNN got 56% AUC, the random forest got 50% and GP with AdaBoost got 63% AUC on the other dataset (cell2cell) KNN got 67% AUC, the random forest got 74% and GP with AdaBoost got 89% AUC. Ahmad et al. (2019) used seriyaTel dataset to build CCP model the dataset contains the record of about 10 million users with 10,000 columns Decision Tree, GBM tree algorithm, XGBoost, and random forest were used for CCP also the best technique to balance the data was an under-sampling technique with the help of this technique XGBoost got 93.12%, GSM got 90.21%, RF got 87.76% and DT got 83%. XGBoost outperformed other models they also used SNA features after adding them XGBoost got the highest accuracy of 93.3%. Fujo et al. (2022) used two datasets one was cell2cell and the other one was the IBM telco dataset they used some machine learning techniques including logistic regression, naive bayes, XGBoost, and KNN models on the other hand they also used deep learning techniques for CCP. Deep-BP-ANN outperforms other models for the cell2cell dataset it got 79.38% accuracy and for the IBM Telco dataset, it got 88.12% accuracy.

CHAPTER 4

Data set

Data is very important for every organization to take better decisions. The dataset utilized for this thesis is Telco customer churn: IBM dataset and Bank Customer Churn Prediction dataset both are available on Kaggle. The dataset contains numerous features that can be helpful to analyze the connection between that features with the churns The telecommunication customers dataset has 7043 rows and 33 columns in this there are few categorical features and few numerical Additionally, the target variable is shown in two columns "Churn Label" contain values (Yes or No) and âChurn Valueâ contains a value (0 or 1) 0 means the customer is not going to be churn and 1 indicates the customer is going to churn any one of these columns can be used as a target variable in this study Churn Value is used as a target variable because machine learning models understand better numerical values so itâs better to use a target variable which has numeric or binary value.

On the other side, The bank customers dataset has 10,000 rows and 14 columns in this there are also a few categorical features and a few are numerical there is also one target variable which is represented in a column âExitedâ containing values (0 or 1). Table 4.1 shows the characteristics of the telecommunication and banking datasets the main core of this thesis is to first build a model to predict the customer churns in telecommunications and then apply the same models in the banking sector. article

Characteristic	Telco Customer	Bank Customer
Total number of customers	7043	10,000
Total number of features	33	14
Missing values	Yes	No
Data distribution	Imbalanced	Imbalanced
Number of churns	1869	2037
Number of not churns	5174	7963
Categorical features	24	3
Numerical features	9	11
Target variable	1	1
Predictor variables	32	13

Table 4.1: Comparison of Characteristics for Telco Customer and Bank Customer Datasets

CHAPTER 5

Performance Evaluation

Any classification model can be evaluated by using accuracy, Precision, Recall, and F1-Score In this study to check the performance of the model I use a confusion matrix. The structure of a confusion matrix is shown in table 5.1.

ACTUAL CLASS			
MODEL PREDICTED CLASS		Customer churn (1)	Not churn (0)
	Customer churn (1)	True positive (TP)	False negative (FN)
	Not churn (0)	False positive (FP)	True negative (TN)

Table 5.1: Structure of confusion matrix

The term True Positive (TP), False Positive (FP), False Negative (FN), and True Negative (TN) are described below:

1. **True Positive (TP):** The True positive occurs when the customer is actually churning and the model also predicts that the customer is churning.

2. **False Positive (FP):** The False positive occurs when the customer is actually not churning and the model predicts that the customer is churning.

3. **False Negative (FN):** The false negative occurs when the customer is actually churning and the model predicts that the customer is not churning.

4. **True Negative (TN):** The true negative occurs when the customer is actually not churning and the model predicts that the customer is not churning.

Accuracy is how accurately the churn prediction model is performed by predicting correct customers who are churn or not this is being calculated with the help of a confusion matrix by the sum of true positive (TP) and true negative (TN) divided by true positive (TP), true negative (TN), false positive (FP) and false negative (FN).

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision is defined as the percentage of the correct predictions out of all predictions made by the machine learning model by true positive (TP) divided by the sum of true positive (TP) and false positive (FP).

$$\text{Precision} = \frac{TP}{TP + FP}$$

The recall is defined as the percentage of the correct predictions out of all the actual results made by the machine learning model by true positive (TP) divided by the sum of true positive (TP) and false negative (FN).

$$\text{Recall} = \frac{TP}{TP + FN}$$

F1-Score tells the performance of the model a high F1 score means the model is performing well.

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

CHAPTER 6

Data Preprocessing

Data pre-processing is one of the most important part of data analysis because data can be collected through various ways like web scrapping, interviews, social surveys, etc. so usually the dataset is not 100% in a proper condition for use to build machine learning models there could be some noisy data in it which can impact the performance of the models and might do wrong predictions so selecting the most suitable preprocessing technique can significantly impact the outcome of the analysis (Engel et al., 2013).

The Telco customer churn: IBM dataset has some issues like missing data, some columns are an object type which has to be in numeric type, etc. On the other hand Bank customer churn dataset has no missing data and every data type was in the correct manner. Before implementing a machine learning model the data should be in a proper manner for that data pre-processing is very important. In this thesis, some data preprocessing techniques are used to enhance the quality of the dataset and to make it usable for creating a machine-learning model the techniques include, handling the missing data, transforming the data using encoding methods to convert categorical columns into numeric did some EDA, scaling the dataset using normalization extract some features to build the ML model moreover both datasets are imbalance so the Smote oversampling technique was used to balance the datasets.

6.1 Handling missing data

Dealing with missing values is a very crucial part because it can affect directly the analysis of the data so it is important to deal with it before analyzing the dataset (Emmanuel et al., 2021). In the Telco customer churn: IBM dataset I have only one column [âTotal Charges â] which has 11 missing values which are quite less in some datasets there are some values that are empty as shown in Nan and can be easily found by using np.nan function but in this case, this Nan values a find by using [â â] this blank space and then convert it into np.nan as this feature has very less missing values otherwise if the column contain 70 to 80% missing values than its better to drop that column there are some techniques to fill the missing data like mean, median, mode, forward imputation, backward imputation but in this study [âTotal chargesâ] column are continues variable so the mean of the complete column was taken to fill those missing values.

6.2 Data transformation

Machine-learning models can better understand numeric data so data transformation can help to increase the performance of the model because in the transformation we convert categorical data into numerical data just like âoffâ and âonâ we convert it into â0â or â1â a dummy variable is also known as an indicator variable, is an additional variable use to represent the categorical feature with two or more unique values. For instance, if there is a column name [Senior Citizen] having values yes or no it converts it into [Senior Citizen_yes] and [Senior Citizen_no] In this thesis one-hot encoding technique is applied to convert categorical features into numerical features by using pandas function pd. get_dummies.

6.3 Feature Selection

Feature selection is a very useful and important process to be done before making a machine learning model. Analyzing data with a high number of rows and columns is very difficult feature selection offers a valuable solution to solve this problem by eliminating the features that are not relevant to the target variable (Cai et al., 2018).

According to (Chandrashekar et al., 2014) feature selection techniques help to reduce the computational time and also enhance the machine learning model prediction performance. Moreover, it also saves our time for instance if we are analyzing 50 features and 25 features are irrelevant by removing those 25 features through the feature selection technique we can reduce almost half of our time and cost.

In this thesis feature selection using elastic-Net regression was used which is a combination of L1 and L2 regularization it offers flexibility to optimize the feature selection process by adjusting the balance between L1 and L2 by using this method I have extracted 7 features from the telecom dataset shown in Figure 6.1.

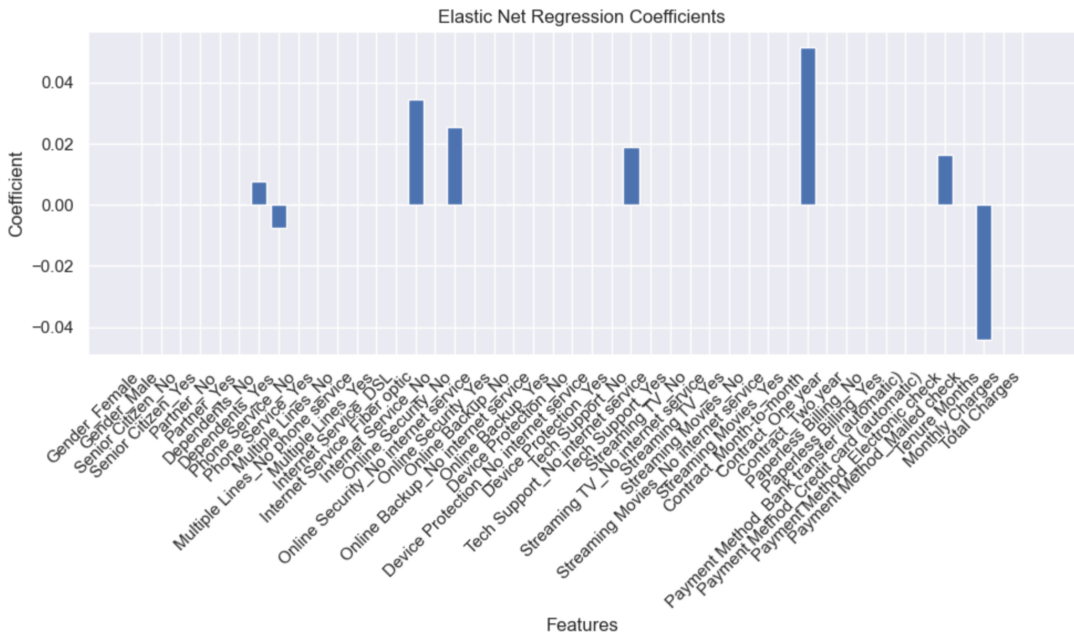

Figure 6.1: Feature selection through Elastic net regression

6.4 Handling Imbalance Data

According to (Batista et al., 2004) there are multiple factors that can affect the performance of the machine learning model among these factors class imbalance data is identified as a significant contributor to affecting the machine learning model performance. Imbalance occurs in classification problems when the target variable is a bit more skewed for instance class â0â has a high occurrence compared to class â1â. In our case we have a target variable as churn 0 means no churns and we have 5174 customers who are not churns on the other hand 1 means churns and we have 1869 customers who are churns so it seems that this is imbalance data.

To tackle the problem of imbalance class, under-sampling, and over-sampling techniques are most frequently used in different domains (Amin et al., 2016). According to (Brownlee, 2020) there are some oversampling techniques that are commonly used which are Random oversampling, (SMOTE) Synthetic minority oversampling technique, (ADASYN) Adaptive synthetic sampling, etc. In this thesis, (SMOTE) Synthetic minority oversampling was used to balance both datasets. Because SMOTE stands out as the most widely utilized technique to solve the class imbalance problem its main objective is to equalize the class distribution by randomly duplicating the minority class (Rout et al., 2023). After applying SMOTE method I got a 5174 ’0’ class value and a 5174 ’1’ class value on the telecom dataset as shown in figure 6.2

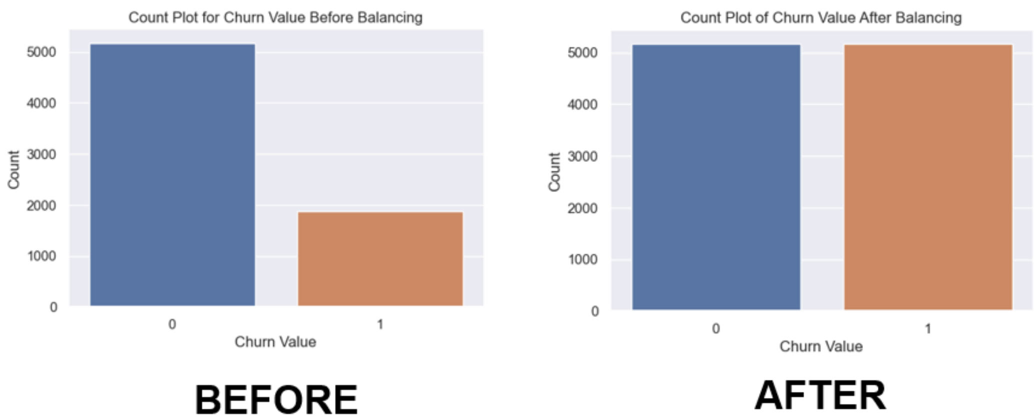

Figure 6.2: comparision of dataset before and after balancing

6.5 Exploratory data analysis (EDA)

1) Churn Rate Analysis with Respect to Gender, Senior Citizen, Partner, and Dependents

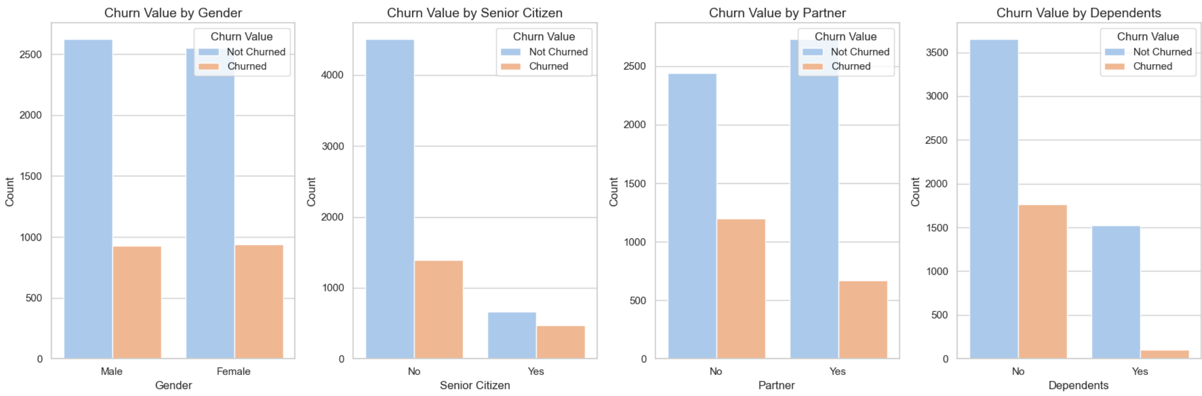

Figure 6.3: churn rates with respect to gender, senior citizen, partner, and dependents

Out of 3555 males, 930 males are experienced as churners on the other hand out of 3488 females 939 are experienced as churners, which shows that the churn rates for both males and females are nearly identical. For senior citizens out of 7043, customers we have 1142 customers who are in the category of a senior citizen, and out of this 476 are churners whereas the count of customers who are not in the category of senior citizens is 5901 out of this 1393 are churners overall it shows that churn rate of a senior citizen is about 41.7% on the other hand churn rate of a non-senior citizen is about 23.6% which is lower than senior citizen rate. For Partners out of 7043 customers we have 3402 customers who have partners and out of this 669 are churners whereas the customers who do not have a partner are 3641 out of these 1200 are churners overall it shows that churn rate of customers who have partners is about 19.7% on the other hand churn rate of customers who do not have a partner is about 33%. For Dependents out of 7043 customers we have 1627 customers who live with any Dependent and out of this 106 are churners whereas the customers who do not have any dependent are 5416 out of this 1763 are churners overall it shows that the churn rate of customers who lives with dependent is about 6.5% on the other hand churn rate of customers who do not live with any dependent is about 32.5%.

Features	Churn Rates	
	Yes	No
Gender	Male(26.1)	Female(26.9)
Senior Citizen	41.7%	23.6%
Partner	19.7%	33%
Dependents	6.5%	32.5%

Table 6.1: Churn Rates with respect to Senior Citizen, Partner, and Dependents

- Customers who are senior citizen has a higher churn rate.
- Customers who do not have partners experienced a higher churn rate.
- Customers who do not have any dependents experienced higher churn rates.

2) Churn Rate Analysis with Respect to Internet Services

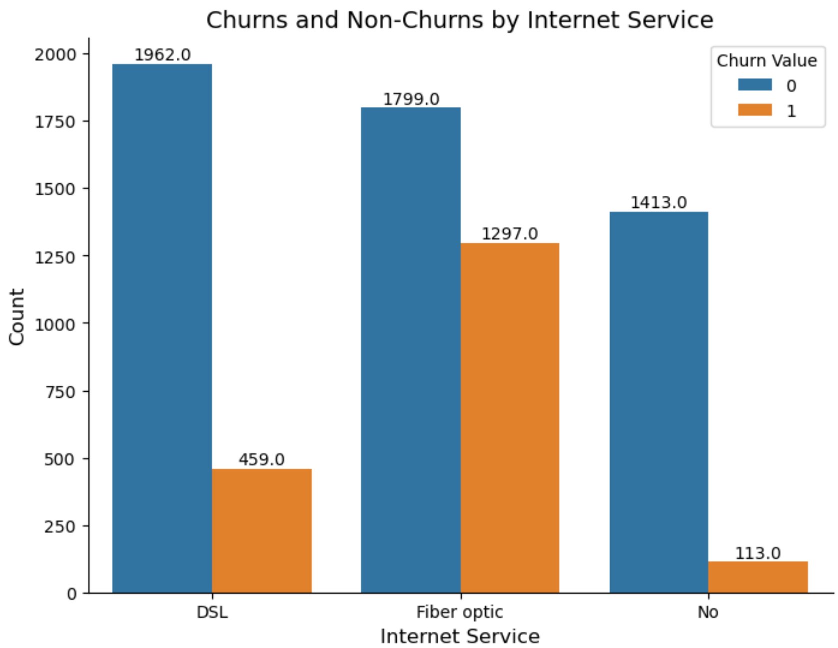

Figure 6.4: churn rates with respect to internet service providers

out of 7043 customers we have 2421 customers who subscribe to DSL internet service and out of this 459 are churners whereas the customers who subscribe to Fiber optic internet service are 3096 out of this 1297 are churners and there are 1526 customers who do not subscribe to any internet service out of this 113 customers are churners. Overall it shows that the churn rate of customers who subscribe to DSL internet service is about 19% on the other hand churn rate of customers who subscribe to Fiber optic internet service is about 41.9%. And the churn rate of customers who do not subscribe to any internet service with the company is about 7.4%. Let's compare the results between two different internet service providers.

a) Fiber Optic Internet Service

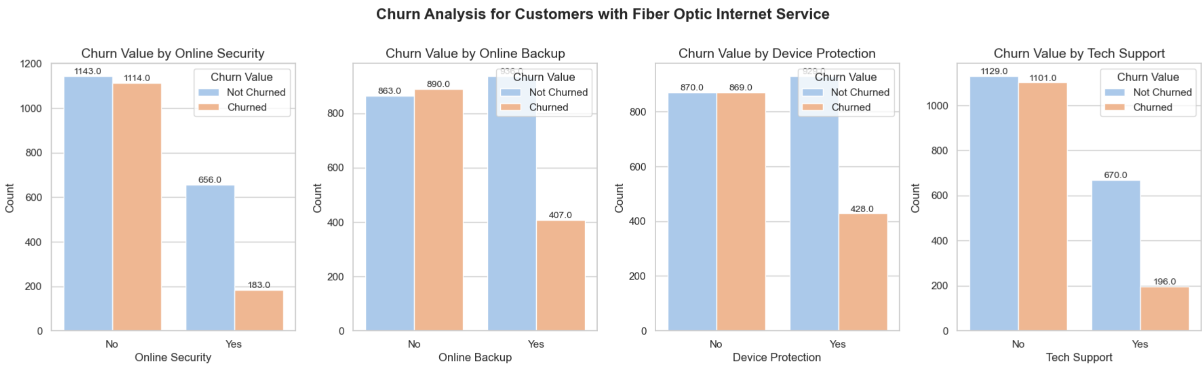

Figure 6.5: customers who are using fiber optic internet service

Out of 3096 customers we have 839 customers who subscribe to an additional online security service and out of this 183 are churners whereas the customers who do not subscribe to an additional online security service are 2257 out of this 1114 are churners. Overall it shows that the churn rate of customers who subscribe to an additional online security service is about 21.8% on the other hand the number of customers who do not subscribe to an additional online security service with the company is about 49.3%. For Online Backup out of 3096 customers we have 1343 customers who subscribe to an additional online backup service and out of this 407 are churners whereas the customers who do not subscribe to an additional online backup service are 1753 out of this 890 are churners. Overall it shows that the churn rate of customers who subscribe to an additional online backup service is about 30.3% on the other hand churn rate of a customer who does not subscribe to an additional online backup service is about 51%. For Device Protection out of 3096 customers we have 1357 customers who subscribe to an additional device protection service and out of this 428 are churners whereas the customers who do not subscribe to an additional device protection service are 1739 out of this 869 are churners. Overall it shows that the churn rate of customers who subscribe to an additional device protection service is about 31.5% on the other hand churn rate of a customer who does not subscribe to an additional device protection service is about 50%. For Tech Support out of 3096 customers we have 866 customers who subscribe to an additional tech support service to reduce wait time and out of this 196 are churners whereas the customers who do not subscribe to an additional tech support service are 2230 out of this 1101 are churners. Overall it shows that the churn rate of customers who subscribe to an additional tech support service is about 22.6% on the other hand churn rate of customers who do not subscribe to an additional tech support service is about 49.3%.

b) DSL Internet Service

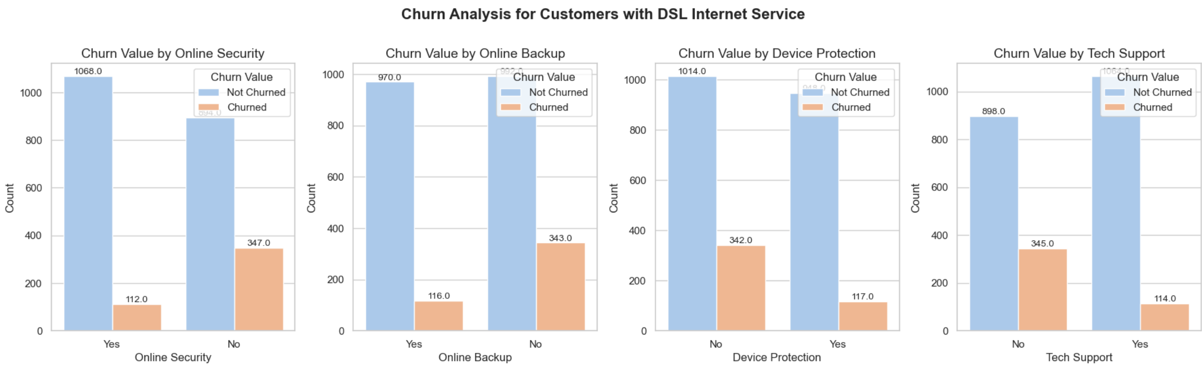

Figure 6.6: customers who are using DSL internet service

Out of 2421 customers we have 1180 customers who subscribe to an additional online security service and out of this 112 are churners whereas the customers who do not subscribe to an additional online security service are 1241 out of this 347 are churners. Overall it shows that the churn rate of customers who subscribe to an additional online security service is about 9.5% on the other hand the number of customers who do not subscribe to an additional online security service with the company is about 28%. For Online Backup out of 2421 customers we have 1086 customers who subscribe to an additional online backup service and out of this 116 are churners whereas the customers who do not subscribe to an additional online backup service are 1335 out of this 343 are churners. Overall it shows that the churn rate of customers who subscribe to an additional online backup service is about 10.7% on the other hand churn rate of a customer who does not subscribe to an additional online backup service is about 25.7%.

For Device Protection out of 2421 customers we have 1065 customers who subscribe to an additional device protection service and out of this 117 are churners whereas the customers who do not subscribe to an additional device protection service are 1356 out of this 342 are churners. Overall it shows that the churn rate of customers who subscribe to an additional device protection service is about 11% on the other hand churn rate of a customer who does not subscribe to an additional device protection service is about 25.2%. For Tech Support out of 2421 customers we have 1178 customers who subscribe to an additional tech support service to reduce wait time and out of this 114 are churners whereas the customers who do not subscribe to an additional tech support service are 1243 out of this 345 are churners. Overall it shows that the churn rate of customers who subscribe to an additional tech support service is about 10% on the other hand churn rate of customers who do not subscribe to an additional tech support service is about 28%.

Services	**Churn Rates**			
	Fiber optic		**DSL**	
Total customers	3096		2421	
	Yes	No	Yes	No
Online security	21.8%	49.3%	9.5%	28%
Online backup	30.3%	51%	10.7%	25.7%
Device protection	31.5%	50%	11%	25.2%
Tech support	22.6%	49.3%	10%	28%

Table 6.2: Churn Rates with respect to different internet service providers with their services

It can be seen that there are more customers who prefer Fiber optic internet service compared to DSL internet service.

- The churn rate indicates that the customers who do not subscribe to an additional online security service have a higher churn rate compared to the customers who subscribe to this online security service.
- The churn rate indicates that the customers who do not subscribe to an additional

online backup service have a higher churn rate compared to the customers who subscribe to this online backup service.

- The churn rates indicate that the customers who do not subscribe to an additional device protection service have a higher churn rate compared to the customers who subscribe for this device protection service.
- The churn rate indicates that the customers who do not subscribe to an additional tech support service have a higher churn rate compared to the customers who subscribe to this tech support service.

3) Churn Rate Analysis with Respect to Contract Type

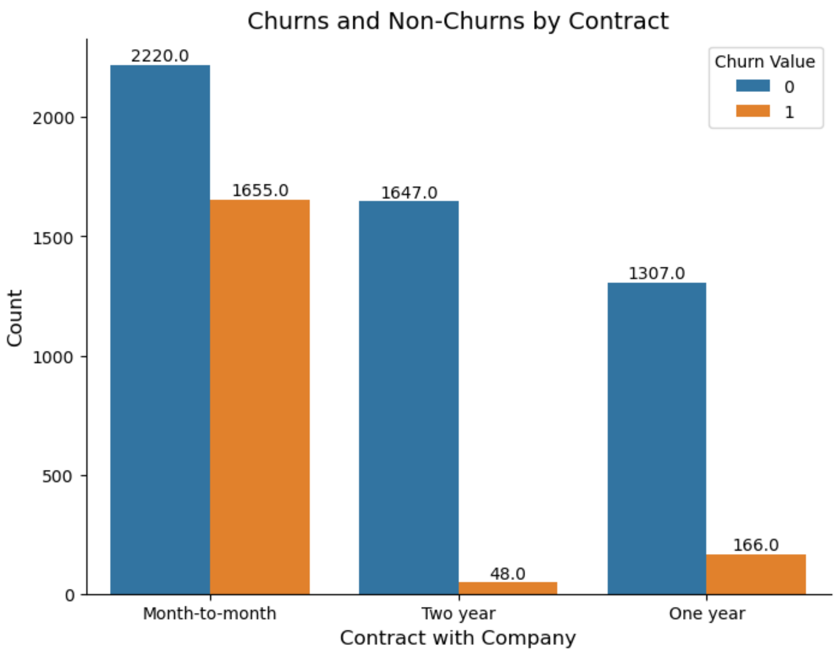

Figure 6.7: churn rates with respect to contract type

Out of 7043 customers we have 3875 customers who use month-to-month contracts and out of this 1655 are churners whereas the customers who use two-year contracts are 1695 out of this 48 are churners and there are 1473 customers who use one-year contracts out of this 166 customers are churners. Overall it shows that the churn rate of customers who use month-to-month contracts is about 42.7% on the other hand churn rate of customers who use two-year contracts is about 2.8%. And the churn rate of customers who use one-year contracts is about 11.3%.

Contract type	**Churn rates**
Month-to-month	42.7%
One-year	11.3%
Two-year	2.8%

Table 6.3: Churn rate with respect to contract type

The churn rates indicate that the customers who use month-to-month contracts have a higher churn rate it also indicates that long-term contracts help to reduce customer churn.

4) Churn Rate Analysis with Respect to Payment Methods

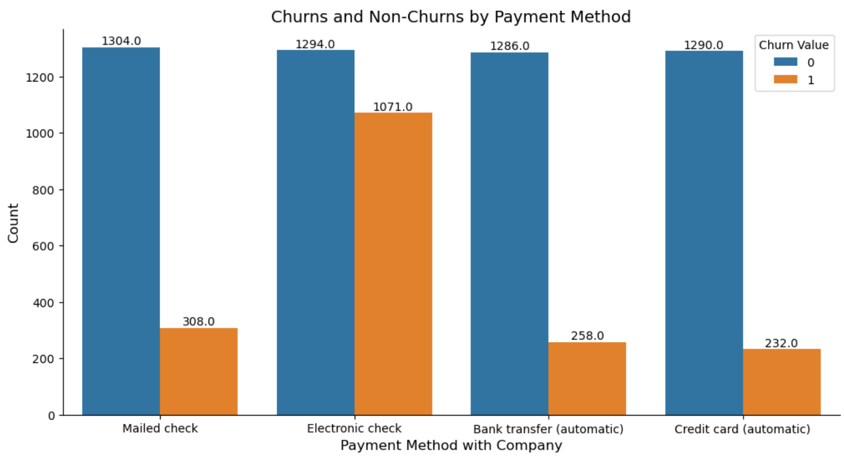

Figure 6.8: churn rates with respect to payment method

Out of 7043 customers, there are 1612 customers who pay their bills by Mailed check, with 308 of them being churners. Additionally, within the group of customers who opt for Electronic check payments, there are 2365 customers, out of which 1071 are churners. Similarly, in the category of customers using Bank transfer (automatic) for bill payments, there are 1544 individuals, with 258 being churners. Furthermore, among the customers who choose Credit card (automatic) payments, we have 1522, and out of these, 232 are churners. Overall it shows that the churn rate of customers who pay their bills by Mailed check is about 19.1%, the churn rate of customers who pay their bills by Electronic check is about 45.3%, the churn rate of customers who pay their bills by Bank transfer (automatic) is about 16.7%, and the churn rate of customers who pay their bills by Credit card (automatic) is about 15.3%

Payment Method	**Churn rates**
Mailed check	19.1%
Electronic check	45.3%
Bank transfer (Automatic)	16.7%
Credit card (Automatic)	15.3%

Table 6.4: Churn rate with respect to payment method

Most customers prefer to pay their bills by Electronic check also they experienced a higher churn rate on the other hand customers who pay their bills by Mailed check, Bank transfer, and Credit card have lower churn rate.

5) Churn Rate Analysis with Respect to Tenure Months

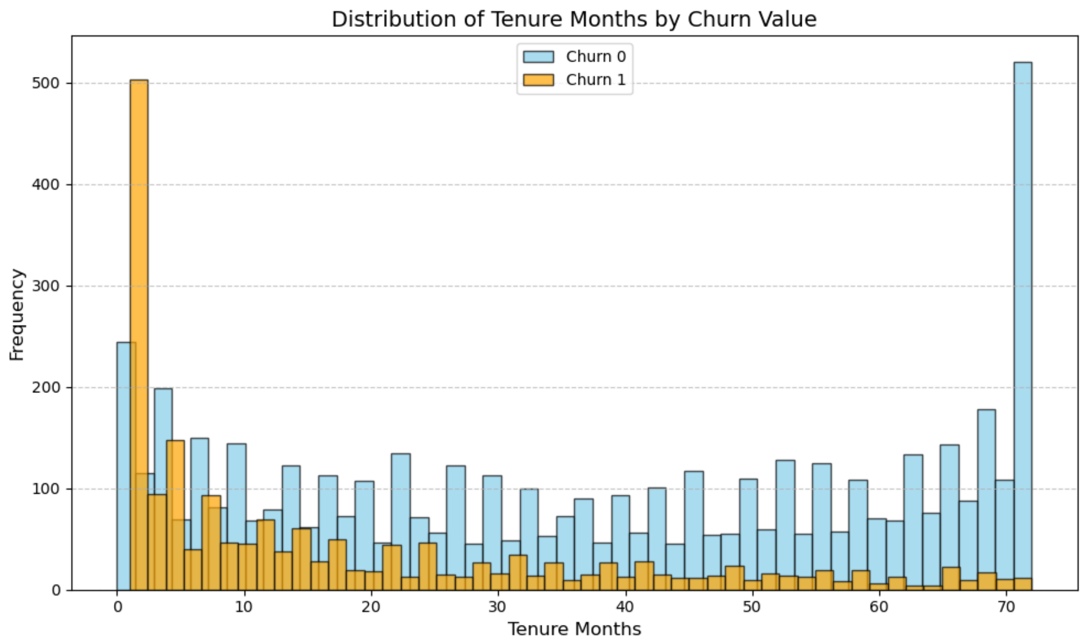

Figure 6.9: churn rates with respect to tenure

The above histogram figure 6.9, shows that the customers with lower tenure months experienced a higher likelihood of churning on the other hand the customers with higher tenure months are less likely to churn which means that if the customer is engaged with the company and getting services for a long time seems to be more loyal and will continue using the services but the customer who just starts using the service needs more attention because the chances to end up the service is very high.

6) Churn Rate Analysis with Respect to Monthly Charges

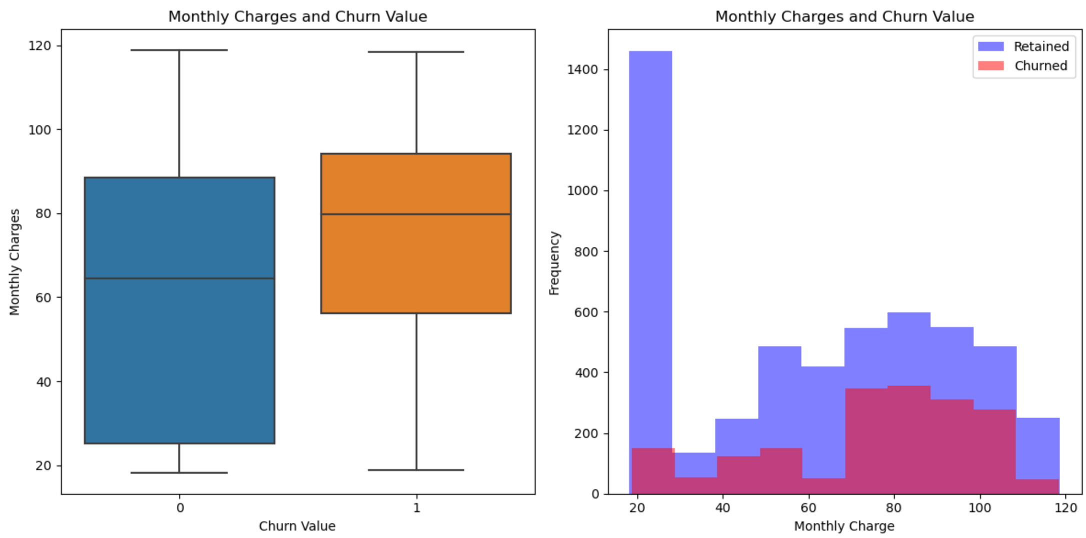

Figure 6.10: churn rates with respect to monthly charges

The above box plot figure 6.10 and histogram show that the average customers who churn tend to have higher monthly charges compared to the customers who do not churn. This means that the monthly charges can play an important role in the churning of customers just like if other companies give the same services for lower charges then might the customer end the relationship with the existing company and switch to another company.

7) Top 10 Cities Which Have Higher Churn Rate

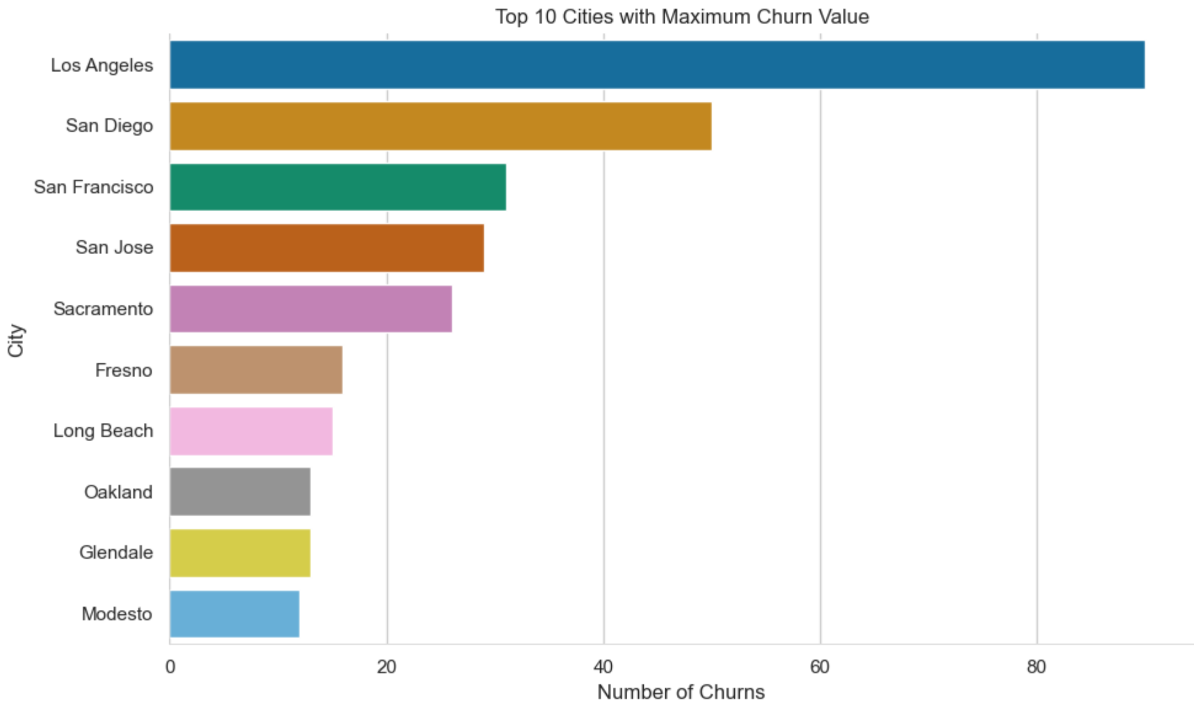

Figure 6.11: Top 10 cities which have highest churn rates

The above bar plot figure 6.11, shows the top 10 cities which has the highest rate of churns and Los Angeles experiencing the most customer churns.

8) Analyzing the Churn Reasons

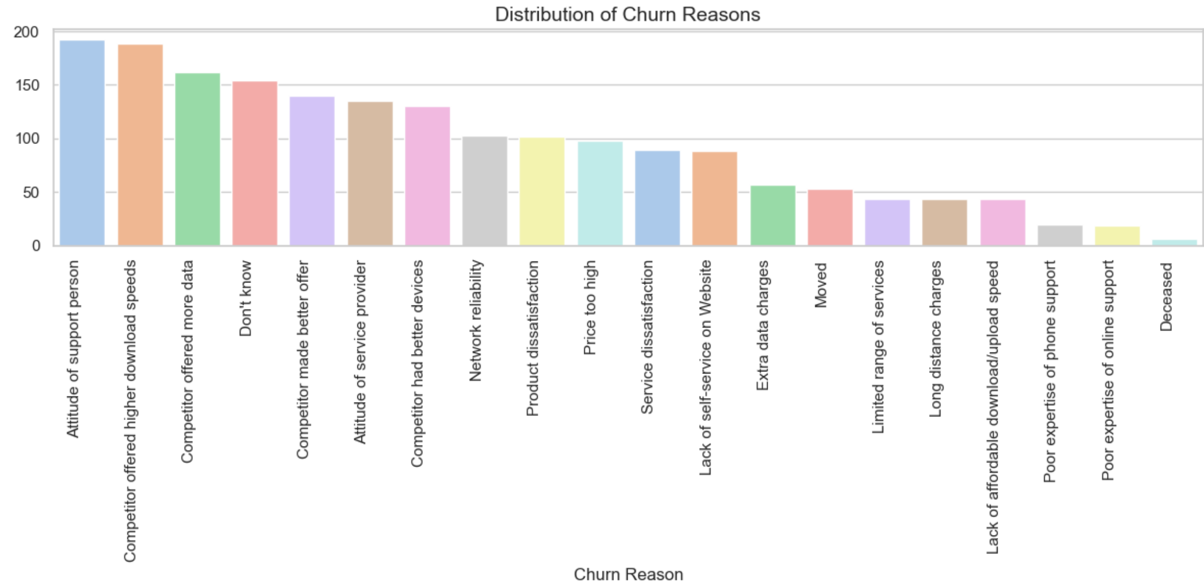

Figure 6.12: customer churn reasons

In the above count plot figure 6.12, it can be seen that the two highest churns reasons are 'Attitude of support person' and the second 'competitor offered higher download speed' if we see the rest of the reasons like 'competitor offered more data' it seems like the services which internet service providers are giving effect more customers to be churn moreover the most important thing for any organization is customer support with an above graph it is also can be seen that most customers are churn because of customer support in terms of 'attitude of support person'.

CHAPTER 7

Methodology

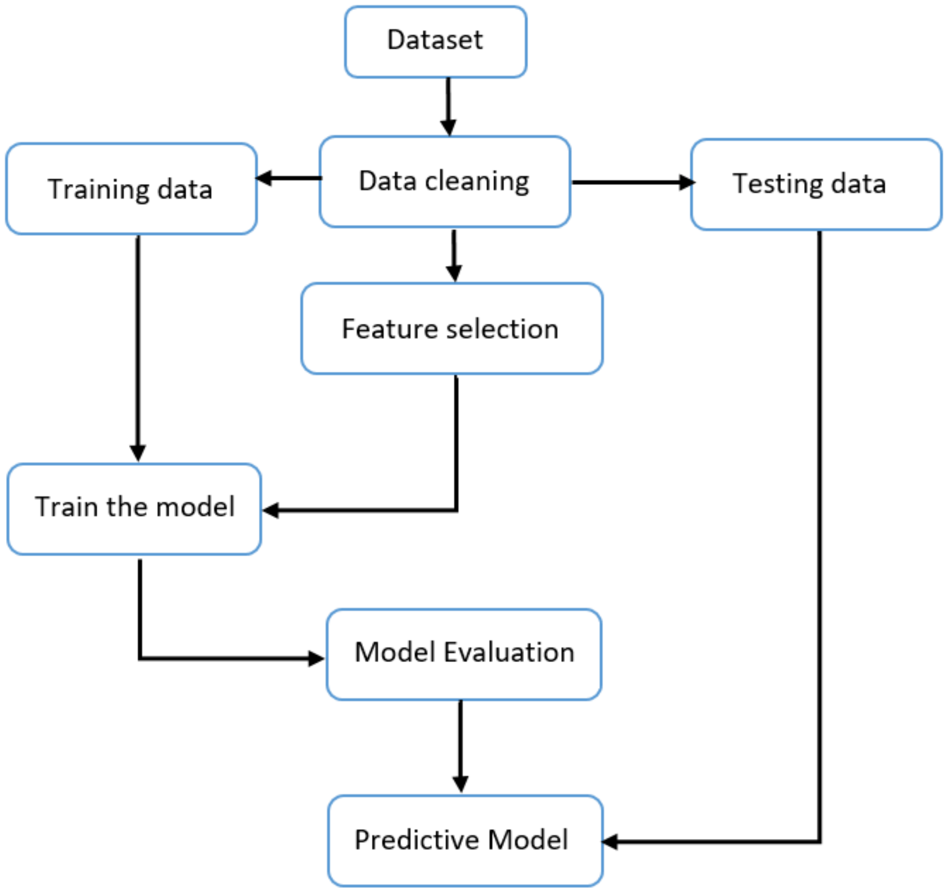

Figure 7.1: Machine learning model cycle

7.1 Training and Testing Sets for ML Models

Training and testing have significant importance in developing any machine learning model training dataset used to train our model for prediction on the other hand the test dataset is the data that gives to the model which it has never seen before and then the machine learning model can be evaluated using that test data. Joseph (2022) Says that the often utilized splitting ratio is 80:20 in which 80% of the data is used for training and 20% of the data is used for testing the model additionally 70:30 in which 70% of the data is used for training and 30% of data is used for testing in some cases 50:50 and 60:40 splitting ratios are used.

In this thesis 80:20 splitting ratio was used by using the train_test_split function with a test_size of 0.2 test size use to define how much data we want for the testing of our model in this case 0.2 means 20% of the data will be used for testing and remaining 80% of data used for training the model. Also using random_stateâ of 42 helps in the consistency of our results which means it will give the same data split every time If we do not use random_state we will get every time random data which can affect the results every time we run the code.

machine learning model cycle is shown in figure 7.1

7.2 Machine Learning For Churn Prediction

In this thesis seven machine learning techniques and two deep learning techniques are used to predict whether the customer is going to leave (churn) or not these methods are as follows:

- Support vector machine (SVM)
- Decision tree
- Naive bayes
- Random forest
- K-Nearest Neighbors (K-NN)
- Logistic regression

- Gradient Boosting
- Artificial Neural Network (ANN)
- Convolutional Neural Network (CNN)

7.2.1 Support Vector Machine (SVM)

Vapnik, Boser, and Guyon introduced the support vector machine (SVM), often known as support vector networks. (Suthaharan et al., 2016). Depending on the dataset, this model may sometimes beat Decision Trees (DT) and Neural Networks (NN) for predicting customer churn (CC) (Shaaban et al., 2012). This model is used to identify the patterns in the data also this is a very popular algorithm for classification and regression analysis I utilize this classification technique to determine whether or not a consumer will leave. A line dividing the data points that maximizes the margin between the hyperplane and the data points is referred to as a hyperplane, and the goal of the supervised machine learning method SVM is to discover the best hyperplane. The data points in this case are known as support vectors.

Model Working:

First the model was created with an instance of the SVM model by using SVC(random_state=42) after that SVM_all.fit was used SVM_all is the model name and fit helped to put the training data into the model standardized X_train_all_fea_std and y_train_all_fea was put. As two models were created one with all features and one with selected features the variable or model containing âallâ refers to all features then â_all.predict was used and put testing data which is X_test_all_fea_std in it to test our model and stored the predictions in SVM_y_pred_allâ variable.

Model Performance Evaluation:

- accuracy_score has been used to measure the accuracy of the model by comparing the predictions (SVM_y_pred_all) with the actual values (y_test_all_fea) and SVM_accuracy_sel is the variable that stores the accuracy of the SVM model.

- precision_score has been used to measure the precision of the model by comparing the predictions (SVM_y_pred_all) with the actual values (y_test_all_fea) and SVM_precision_all is the variable that stores the precision score of the SVM model.

- recall_score has been used to measure the recall score of the model by comparing the predictions (_y_pred_all) with the actual values (y_test_all_fea) and SVM_recall_all is the variable that stores the recall score of the SVM model.

- f1_score has been used to measure the F1-score of the model by comparing the predictions (SVM_y_pred_all) with the actual values (y_test_all_fea) and SVM_recall_all is the variable that stores the F1-score of the SVM model.

The 10-fold cross-validation was also performed by using the cross_val_score function inside that function the model name, and training data(X and y) were putted, to use 10 fold-cross validation cv=10 was used which will make 10 subsets of the training dataset, and put âaccuracyâ in scoring parameter to define that the evaluation metrics is accuracy and SVM_cv_scores_all_features stores the list of accuracy obtained by each subset of training data.

All the above steps have been done for the model with selected features.

7.2.2 Decision Tree

This model was also used in this thesis to forecast whether the consumer will go or stay because decision trees are really useful and have been immensely popular for dealing with classification issues. A decision tree is more similar to common trees they have branches, leaves, and roots the same representation is applied to decision tree models it consists of the root node which is a starting point each node represents the specific features and each branch indicates the decisions or rule at last the leaf node show the end result (Patel et al., 2018). Decision trees might have some branches that arise because of noise in the data the tree pruning process helps to eliminate those branches and increase the accuracy of the model (Au et al., 2003).

Model Working:

First, the instance of the Decision Tree model was created by using DecisionTreeClassifier(random_state=42) after that the same step was followed that is mentioned in the first model working.

7.2.3 Naive Bayes

Naive bayes algorithm is a simple probabilistic classifier that uses bayes theorem these classifiers are built on the assumption that the effect of each feature has an independent influence on a given class The term naive is used in naive bayes because we make a naive assumption that features such as gender, senior citizen, partner, dependents, etc. are independent of each other. Which eases the calculations and improves the algorithmâs computational speed (Dong et al., 2011). This algorithm might not perform well on a big dataset but in our case, we have in total (7043 rows and 33 columns) so in this thesis naive bayes algorithm was used to predict the CC (Customer Churn) and it displayed satisfactory results.

Model Working:

First, the instance of the naive bayes model was created by using the GaussianNB function after that the same step was followed that is mentioned in the first model working.

7.2.4 Random Forest Classifier

Random forest is one of the powerful algorithm of machine learning to perform classification tasks for predicting whether the customer is planning to end their subscription with the company or not in this thesis a random forest classifier is also used. The process involves dividing one dataset into many sample datasets and for each set, it makes a separate decision tree and gets an output from all trees, and based on majority votes it gives us one result. In comparison to traditional algorithms, random forest has good qualities and advantages which is why it is been used widely for classification and prediction tasks (Liu et al., 2012).

Model Working:

First, the instance of the Random Forest model was created by using the RandomForestClassifier(random_state=42) function after that the same step was followed that is mentioned in the first model working.

7.2.5 K-Nearest Neighbors (K-NN)

KNN works as a majority voting system where majority class labels decide what would be the new class label (Jodas et al., 2023). For instance, you are in the garden and have a bucket filled with fruits like mango, apple and suddenly you come across a fruit that you donât know what is this so what you can do is see 5 nearest trees these trees are your neighbors if most trees have apple then the fruit you found is an apple. It is also known as a lazy algorithm it can be useful for small datasets so in this case to predict the CC In this thesis this model was also used.

Model Working:

First, the instance of the K-Nearest Neighbors (k-NN) model was created by using the KNeighborsClassifier(n_neighbors=5) function after that the same step was followed that is mentioned in the first model working.

7.2.6 Logistic Regression

Let’s say you want to predict whether a fruit is an apple or a mango based on taste, color, and size Logistic regression helps you to give the probability of which fruit it belongs to. Logistic regression performs well in binary classification problems as in our situation we need to predict CC which is a binary problem 1 means the customer is going to leave the company and 0 means the customer will not leave so this logistic regression model was also used and it gives satisfactory results.

Model Working:

First, the instance of the Logistic Regression model was created by using the LogisticRegression(random_state=42) function after that the same step was followed that is mentioned in the first model working.

7.2.7 Gradient Boosting

Machine learning algorithms are divided into Bias error and variance error gradient boosting algorithm helps to reduce the bias error moreover gradient boosting is not only limited to predicting continuous values it can also be able to predict categorical values (Bentejac et al., 2021). In simple words, it can be used for both to predict numbers or labels.

Model Working:

First, the instance of the gradient boosting model was created by using the GradientBoostingClassifier(random_state=42) function after that the same step was followed that is mentioned in the first model working.

7.2.8 Artificial Neural Network (ANN)

A neural network works in a sequence of layers in which each layer does a specific task to make a neural network model kerasṠequential was used after that kerasİlayersḊense function was used to create a dense layer which is used to make a connection between neurons. The first layer contains 46 neurons and input_shape=(46,) means there are in total 46 input features I used Rectified Linear Unit (ReLU) as an activation function. The next layer is the output layer and has only 1 neuron because we need to predict the CC in which we have only two values 0 and 1 hence it is a classification problem so for that sigmoid as an activation function was used. For the model compilation, the model.compile method was used in this method there were three parameters first one is the optimizer which is used to update the weights of the input features during model training second parameter is the âlossâ function in which I use binary_crossentropy because this is a binary classification problem. The third parameter is metrics in which accuracy was used as an evaluation matrix. After that, fit function was used to put the training set in it and there are 60 iterations (epochs=60) for which the model is trained. By using the âevaluateâ function the accuracy of the model has been evaluated.

Then the test data passed to the model and use the âpredictâ function and store the predictions in the y_predâ variable. (y_pred > 0.5) was also used as the threshold for the prediction which means that if the predicted probabilities are less than 0.5 means it is a wrong prediction and if it is greater than 0.5 means it is a correct prediction.

7.2.9 Convolutional Neural Network (CNN)

Firstly the instance of the model was created by using the âSequential()â function than two layers are added first is a 1D convolutional layer that contains 64 filters, kernel size is 3, and relu is used as an activation function the second layer is 1D max pooling layer with a pool size of 2 after than flatten layer is added by using add(Flatten()) function it converts the 2D output to 1D vector after that 2 fully connected layers are dense added the first layer has 46 neurons with relu as an activation function and the second layer has 1 neuron and used sigmoid as an activation function for the compilation the same parameters which were used in ANN was used. After that, the fit function was used to put the training set in it and there are 55 iterations (epochs=55) for which the model is trained then for evaluation of the model same steps were taken as in ANN.

CHAPTER 8

Results and Findings

8.1 Telecommunication Customer Dataset

In the Telco customer churn: IBM dataset we have records of 7043 customers out of this 26.5% of customers are churners and 73.5% remain loyal to the company which is shown in figure 8.1. After doing feature selection using Elastic Net Regression we have got these columns ['Dependents_No', 'Dependents_Yes', 'Internet Service_Fiber optic', 'Online Security_No', 'Tech Support_No', 'Contract_Month-to-month', 'Payment Method_Electronic check', 'Tenure Months'] Monthly Charges column is also can be the very important feature so this feature is also added into selected features. This dataset is imbalanced so first to balanced the dataset SMOTE technique was used after that two models of each were built, one with all features and one with selected features to check the performance of the models with all and selected features.

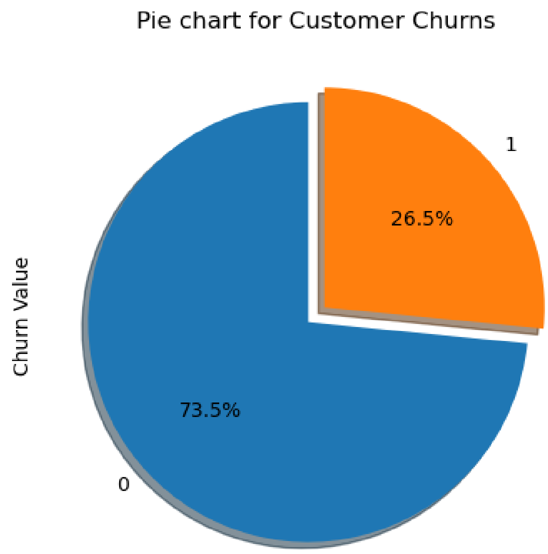

Figure 8.1: churn rates in telecommunication customer dataset

The Logistic Regression model with all features achieved an accuracy of 86% and 88% precision value, 83% Recall, and F1-Score stood at 86% on the other hand, the Logistic Regression model with selected features achieved an accuracy of 77% and 77% precision value, 78% Recall, and F1-Score stood at 77%. The Naive Bayes model with all features achieved an accuracy of 81% and 78% precision value, 86% Recall, and F1-Score stood at 82% on the other hand, the Naive Bayes model with selected features achieved an accuracy of 74% and 73% precision value, 77% Recall, and F1-Score stood at 75%. The Decision Tree model with all features achieved an accuracy of 81% and 82% precision value, 80% Recall, and F1-Score stood at 81% on the other hand, the Decision Tree model with selected features achieved an accuracy of 78% and 79% precision value, 78% Recall, and F1-Score stood at 79%. The Random Forest model with all features achieved an accuracy of 86% and 87% precision value, 85% Recall, and F1-Score stood at 86% on the other hand, the Random Forest model with selected features achieved an accuracy of 81% and 81% precision value, 82% Recall, and F1-Score stood at 81%. The Support Vector Machines (SVM) model with all features achieved an accuracy of 85% and 88% precision value, 82% Recall, and F1-Score stood at 85% on the other hand, the Support Vector Machines (SVM) model with selected features achieved an accuracy of 78% and 77% precision value, 82% Recall, and F1-Score stood at 79%. The K-Nearest Neighbors (k-NN) model with all features achieved an accuracy of 83% and 83% precision value, 83% Recall, and F1-Score stood at 83% on the other hand, the K-Nearest Neighbors (k-NN) model with selected features achieved an accuracy of 78% and 77% precision value, 83% Recall, and F1-Score stood at 80%. The Gradient Boosting model with all features achieved an accuracy of 87% and 87% precision value, 87% Recall, and F1-Score stood at 87% on the other hand, the Gradient Boosting model with selected features achieved an accuracy of 80% and 77% precision value, 85% Recall, and F1-Score stood at 81%.

When 10-fold cross-validation was applied to all models Logistic regression with all features got 84% accuracy and with selected features, it got 76%. Naive Bayes with all features got 80% accuracy and with selected features, it got 73%. Decision Tree with all features got 81% accuracy and with selected features, it got 78%. Random Forest with all features got 86% accuracy and with selected features, it got 80%. Support Vector Machines (SVM) with all features got 65% accuracy and with selected features, it got 73%. K-Nearest Neighbors (k-NN) with all features got 76% accuracy and with selected features, it got 78%. Gradient Boosting with all features got 86% accuracy and with selected features, it got 78%.

The Artificial Neural Network (ANN) with 46 neurons in the first layer with Relu activation and in the second layer 1 neuron with sigmoid activation with 55 epochs achieved an accuracy of 89.42% and 87% precision value, 84% recall, and F1-Score stood at 85%.The Convolutional Neural Network (CNN) with 55 epochs achieved an accuracy of 89% and 86% precision value, 83% recall, and F1-Score stood at 85%.

ML Models	**Cross-Validation Accuracy**	
	All features	**Selected features**
Logistic Regression	84	76
Naive Bayes	80	73
Decision Tree	81	78
Random forest	86	80
Support vector machine (SVM)	65	73
K-nearest neighbors (K-NN)	76	78
Gradient boosting	86	78
Deep Learning Models Accuracy		
Artificial neural network (ANN)	89.42	
Convolutional neural network (CNN)	89	

Table 8.1: Machine learning and deep learning model performance (Telecom)

In the Telecom customer dataset, in table 8.1 it can be seen that only the Support vector machine (SVM) model with selected features performs better otherwise all models with all features perform better than the models that were made with selected features In machine learning Gradient Boosting and Random forest models perform better but in terms of accuracy Gradient Boosting perform slightly better. In deep learning, almost both models outperform machine learning models but the artificial neural network (ANN) slightly performs better.

The comparison between the best machine learning (with all faetures) and deep learning models performance is shown in figure 8.2.

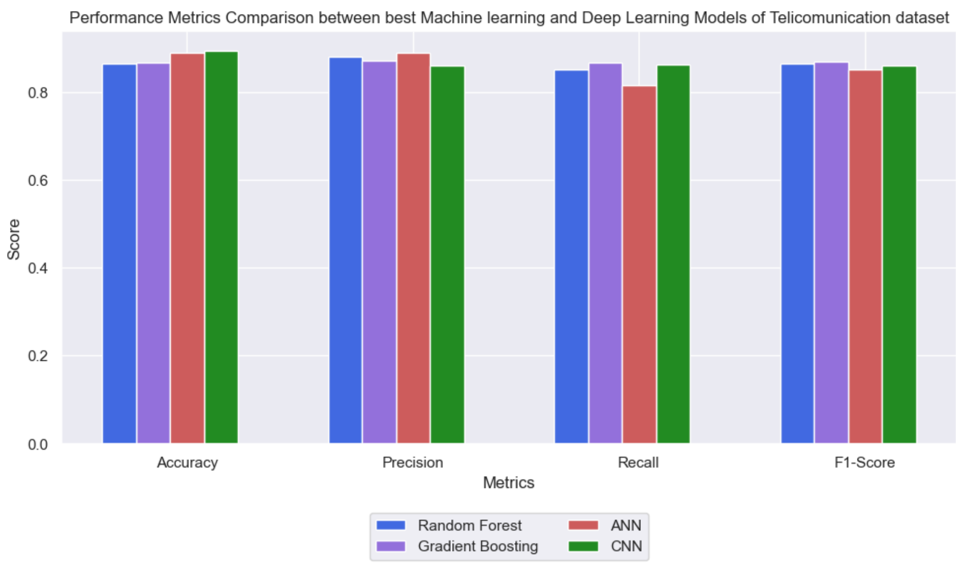

Figure 8.2: Best machine learning (with all features) and deep learning models performance

8.2 Bank Customer Dataset

In the Bank Customer Churn Prediction dataset, we have records of 10,000 customers out of this 20.4% of customers are churners, and 79.6% remain loyal to the company which is shown in figure 8.3. All features had been selected because in the Telecommunications dataset machine learning models with all features seem to be performed better than the models with selected features. This dataset was also imbalanced so the same technique was applied to balanced the dataset using SMOTE and then applied machine learning models.

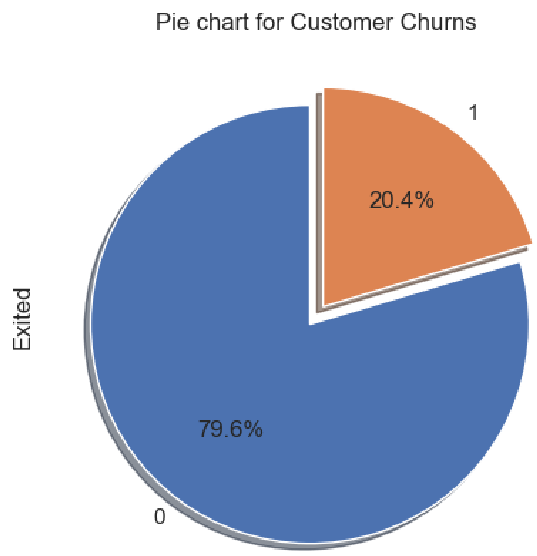

Figure 8.3: churn rates in bank customer dataset

The Logistic Regression model with all features achieved an accuracy of 84% and 88% precision value, 78% Recall, and F1-Score stood at 83%. The Naive Bayes model with all features achieved an accuracy of 80% and 80% precision value, 78% Recall, and F1-Score stood at 79%. The Decision Tree model with all features achieved an accuracy of 84% and 83% precision value, 86% Recall, and F1-Score stood at 84%. The Random Forest model with all features achieved an accuracy of 90% and 91% precision value, 88% Recall and F1-Score stood at 89%. The Support Vector Machines (SVM) model with all features achieved an accuracy of 89% and 92% precision value, 84% Recall, and F1-Score stood at 88%. The K-Nearest Neighbors (k-NN) model with all features achieved an accuracy of 87% and 88% precision value, 84% Recall, and F1-Score stood at 86%. The Gradient Boosting model with all features achieved an accuracy of 89% and 91% precision value, 85% Recall, and F1-Score stood at 88%.

When 10-fold cross-validation was applied to all models Logistic regression got 69% accuracy. Naive Bayes got 72% accuracy. The decision Tree got 84% accuracy. Random Forest got 89% accuracy. Support Vector Machines (SVM) got 58% accuracy. K-Nearest Neighbors (k-NN) got 68%. Gradient Boosting got 88% accuracy.

The Artificial Neural Network (ANN) with 13 neurons in the first layer with Relu activation and in the second layer 1 neuron with sigmoid activation with 60 epochs achieved an accuracy of 88% and 90% precision value, 86% recall, and F1-Score stood at 88%. The Convolutional Neural Network (CNN) with 55 epochs achieved an accuracy of 90% and 90% precision value, 85% recall, and F1-Score stood at 88%.

The comparison between the best machine learning and deep learning models performance is shown in figure 8.4.

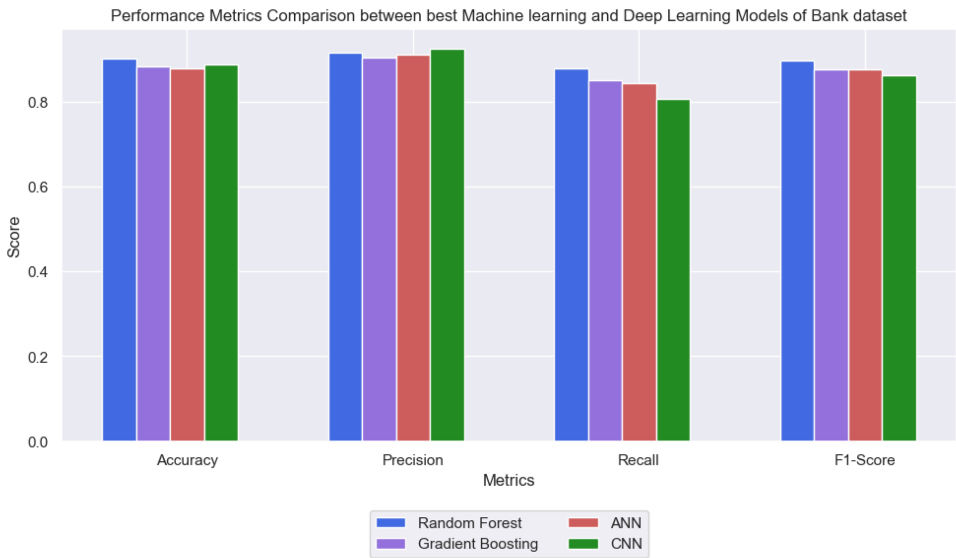

Figure 8.4: Best machine learning and deep learning models performance

ML Models	Cross-Validation Accuracy
Logistic Regression	69
Naive Bayes	72
Decision Tree	84
Random forest	89
Support vector machine (SVM)	58
K-nearest neighbors (K-NN)	68
Gradient boosting	88
Deep Learning Models Accuracy	
Artificial neural network (ANN)	88
Convolutional neural network (CNN)	90

Table 8.2: Machine learning and deep learning model performance (bank)

In the Bank customer dataset, in table 8.2 it can be seen that the Support vector machine (SVM) model performs worse whereas Random Forest and Gradient Boosting perform better than other models moreover Random Forest performs slightly better. In deep learning, both models perform very well but Convolutional Neural Network (CNN) performs slightly better.

CHAPTER 9

Conclusions

Nowadays the competition between every organization in every sector is at its peak everyone wants to retain their existing customers and they want to keep their customers loyal to the organization and try their best to make them not end up a relationship with the company. There are two major sectors that are highly affected by customer churn one is the telecommunication sector and the other one is banking sector In this study we tried to use machine learning techniques to make models that can predict whether the customer is going to leave or not moreover we mainly focused to analyze telecommunication sector and made predictive models for that after that we want to check the transferability of the machine learning models like the model which work best in telecommunication do this same model can give the similar results in other sectors?

In this thesis, a few machine learning and deep learning models including logistic regression, Naive Bayes, decision trees, random forest, support vector machine (SVM), K-nearest neighbor (KNN), gradient boosting, rrtificial neural networks (ANN), Convolutional neural network (CNN) have been applied.

For the Telecommunication sector, the Telco customer churn: IBM dataset has been used containing 7043 customers from which 1869 customers are churners and 5174 customers are non-churners on the other hand for the Banking sector âBank Customer Churn Prediction datasetâ containing 10,000 customers records from which 2037 are churners and 7963 are non-churners as our primary goal is to make a model for telecommunication for that feature selection using elastic net regression was used and got 8 features but one âmonthly chargesâ feature is also very important so this feature was also added in selected features too.

Both datasets are imbalanced so smote technique has been applied to balance the datasets to see the comparative analysis of which model performs better two of each model was created one model with all features and one with selected feature and the models with all features produce good results.

The two best machine learning models in Telecom dataset which performs best amongst other are Random forest which got 87% accuracy, with 88% precision, 85% recall, and 87% F-1 score. The other one is Gradient Boosting which got 87% accuracy with 87% precision, 86% recall and 87% F-1 score. With the 10-fold cross-validation Random forest got 86% accuracy and gradient boosting got 87% accuracy. Moreover, in deep learning, all three models perform almost the same but Artificial Neural Network (ANN) got the highest accuracy of 90%.

This thesis aims to check the transferability of the machine learning models for which one model to predict whether the customer is going to churn or not was built and used in the different sectors just like in the banking sector so in the telecom dataset the models with all features produce good results by considering that in banking dataset the model was built with all features and The two best machine learning models in banking dataset which outperformed other models are Random forest which got 90% accuracy, with 92% precision, 87% recall, and 89% F-1 score. The other one is Gradient Boosting which got 89% accuracy with 91% precision, 85% recall and 88% F-1 score. With the 10-fold cross-validation, Random forest got 89% accuracy and gradient boosting got 88% accuracy. Moreover, in deep learning all three models give almost the same results but Convolutional Neural Network (CNN) got the highest accuracy of 90%.

It can be seen that in both sectors machine learning models including Random Forest and Gradient Boosting produces good results with almost the same accuracy and in deep learning all models produce similar results but Artificial Neural Network (ANN) can be used in both sectors to predict the CC. Over time we might need to make some changes in our models depending on the features it is just like if you make one recipe of sweets you can't exactly apply the same recipe for all sweets, there might be some ingredients that could be different in a similar way telecom and banking sectors are having different features but there are few features that are almost same which both organizations consider but in future, there could be some more features coming up in both sectors depending on humans behavior so over time both industries might need to do some alteration in their models to run it in a smooth way. Roughly there is a high possibility that these two machine learning models and one deep learning model which are mentioned above will produce the best results in all sectors having the same CC problem.

CHAPTER 10

Recommendation

Nowadays there is a huge amount of data is being generated by human beings meanwhile in the telecommunication industry the number of users is very high because these companies provide an easy way of communicating with each other so it is a fact that the amount of data these companies are generating is very high and machine learning models do not perform well in the big dataset for that deep learning models can play a major role to deal with a large data In this study we found that the artificial neural network (ANN) and convolutional neural network (CNN) models can be helpful to predict whether the customer is going to leave or not but if the dataset is short just like if a new startup wants to predict the CC and they donât have millions of customer base so the use of machine learning models includes random forest and gradient boosting would be the best choice.

In this thesis the dataset used does not have enough features there could be more features added just as calling time, which package customers are more likely to subscribe to, feedbacks, and monthly complaints, in the family do all members use the same network? And monthly income etc. By using more relevant features we can make our models more effective to make good predictions hence telecommunication industry needs to get this data from their customers.

By analyzing the data there are two main reasons customers end their relationship with companies the first is bad customer service in the modern era the key to success in every business is to provide good customer service to customers. Time is very important for everyone to solve customer problems efficiently, reducing delays saves the customer time and makes them happy, and gives them a reason to remain loyal to the company the second most important reason is the other company giving the best offers so it is necessary for telecommunication companies to keep their eyes on other companies what they are offering to the customers because if they offer better service on a cheaper rate then there is a high chance that customer will leave the existing company and join other who offer good packages so companies should try to give their customers good packages by the time introduce some discounts on special occasions. They should try to convince customers to take a long-term plan with good offers because in the data I analyzed the customers having long-term subscriptions are less likely to churn, also There are more customers who use electronic checks as a payment method but they are the most experienced as churners so the company should shift their customers to use mailed check, bank transfer, and credit cards as a payment method. Moreover, companies need to resolve the issues that customer is facing while using this internet service because the data shows that more customers prefer to use fiber optic service meanwhile the customers using this service experience more churners company also need to encourage customers to use online security, device protection, online backup, and tech support services because of that customers can get proper benefits which give them a reason to remain loyal with the company.

Chapter 11

Future Work

Data from various nations will be utilized to determine whether any characteristics shared by all nations are significant for CC also employ a few additional features that can assist in strengthening and improving our machine-learning model.

Compare this machine learning CC predictive models in different sectors like banking, hospital, and marketing industry, and also implement some more deep learning techniques like ensemble learning, and transfer learning.

Implement this machine learning model with web applications and deploy it using AWS. Our system will automatically identify the customers who will be likely to churn and take company feedback to improve this model over time.

Bibliography

[1] Amin, A., Anwar, S., Adnan, A., Nawaz, M., Howard, N., Qadir, J., Hussain, A. (2016). Comparing oversampling techniques to handle the class imbalance problem: A customer churn prediction case study. Ieee Access, 4, 7940-7957.

[2] Almuqren, L., Alrayes, F. S., Cristea, A. I. (2021). An Empirical Study on Customer Churn Behaviours Prediction Using Arabic Twitter Mining Approach. Future Internet, 13(7), Article 175.

[3] Ahmad, A. K., Jafar, A., Aljoumaa, K. (2019). Customer churn prediction in telecom using machine learning in big data platform. Journal of Big Data, 6(1), 1-24.

[4] Ahmad, Abdelrahim Kasem, Assef Jafar, and Kadan Aljoumaa. "Customer churn prediction in telecom using machine learning in big data platform." Journal of Big Data 6, no. 1 (2019): 1-24

[5] Au, W. H., Chan, K. C., Yao, X. (2003). A novel evolutionary data mining algorithm with applications to churn prediction. IEEE transactions on evolutionary computation, 7(6), 532-545.

[6] Batista, G. E., Prati, R. C., Monard, M. C. (2004). A study of the behavior of several methods for balancing machine learning training data. ACM SIGKDD explorations newsletter, 6(1), 20-29.

[7] Bentejac, C., Csorgo, A., Martinez-Munoz, G. (2021). A comparative analysis of gradient boosting algorithms. Artificial Intelligence Review, 54, 1937-1967.

[8] Bhattacharyya, J., Dash, M. K. (2022). What do we know about customer churn behaviour in the telecommunication industry? A bibliometric analysis of research trends, 1985â2019. FIIB Business Review, 11(3), 280-302.

[9] Brownlee, J. (2020). Tour of data sampling methods for imbalanced classification. Machine Learning Mastery. Retrieved from https://machinelearningmastery. com/data-sampling-methods-for-imbalancedclassification.

[10] Cai, J., Luo, J., Wang, S., Yang, S. (2018). Feature selection in machine learning: A new perspective. Neurocomputing, 300, 70-79.

[11] Chandrashekar, G., Sahin, F. (2014). A survey on feature selection methods. Computers Electrical Engineering, 40(1), 16-28.

[12] Dong, T., Shang, W., Zhu, H. (2011, May). Naive bayesian classifier based on the improved feature weighting algorithm. In International Conference on Computer Science and Information Engineering (pp. 142-147). Berlin, Heidelberg: Springer Berlin Heidelberg.

[13] Emmanuel, T., Maupong, T., Mpoeleng, D., Semong, T., Mphago, B., Tabona, O. (2021). A survey on missing data in machine learning. Journal of Big Data, 8(1), 1-37.

[14] Engel, J., Gerretzen, J., SzymaÅska, E., Jansen, J. J., Downey, G., Blanchet, L., Buydens, L. M. (2013). Breaking with trends in pre-processing?. TrAC Trends in Analytical Chemistry, 50, 96-106.

[15] Fujo, S. W., Subramanian, S., Khder, M. A. (2022). Customer churn prediction in telecommunication industry using deep learning. Information Sciences Letters, 11(1), 24.

[16] Huang, B., Kechadi, M. T., Buckley, B. (2012). Customer churn prediction in telecommunications. Expert Systems with Applications, 39(1), 1414-1425.

[17] Idris, A., Khan, A., Lee, Y. S. (2012, October). Genetic programming and adaboosting based churn prediction for telecom. In 2012 IEEE international conference on Systems, Man, and Cybernetics (SMC) (pp. 1328-1332). IEEE.

[18] Jain, H., Khunteta, A., Srivastava, S. (2020). Churn prediction in telecommunication using logistic regression and logit boost. Procedia Computer Science, 167, 101-112.

[19] Janiesch, Christian, Patrick Zschech, and Kai Heinrich. "Machine learning and deep learning." Electronic Markets 31, no. 3 (2021): 685-695.

[20] Jodas, D. S., Passos, L. A., Adeel, A., Papa, J. P. (2023). PL-kNN: A Python-based implementation of a parameterless k-Nearest Neighbors classifier. Software Impacts, 15, 100459.

[21] Joseph, V. R. (2022). Optimal ratio for data splitting. Statistical Analysis and Data Mining: The ASA Data Science Journal, 15(4), 531-538.

[22] Kim, M. K., Park, M. C., Jeong, D. H. (2004). The effects of customer satisfaction and switching barrier on customer loyalty in Korean mobile telecommunication services. Telecommunications policy, 28(2), 145-159.

[23] Lalwani, P., Mishra, M. K., Chadha, J. S., Sethi, P. (2022). Customer churn prediction system: a machine learning approach. Computing, 1-24.

[24] Liu, Y., Wang, Y., Zhang, J. (2012). New machine learning algorithm: Random forest. In Information Computing and Applications: Third International Conference, ICICA 2012, Chengde, China, September 14-16, 2012. Proceedings 3 (pp. 246-252). Springer Berlin Heidelberg.

[25] Patel, H. H., Prajapati, P. (2018). Study and analysis of decision tree based classification algorithms. International Journal of Computer Sciences and Engineering, 6(10), 74-78.

[26] Rout, N., Mishra, D., Mallick, M. K. (2023). An advance extended binomial GLMBoost ensemble method with synthetic minority over-sampling technique for handling imbalanced datasets. International Journal of Electrical and Computer Engineering (IJECE), 13(4), 4357-4368.

[27] Suthaharan, S., Suthaharan, S. (2016). Support vector machine. Machine learning models and algorithms for big data classification: thinking with examples for effective learning, 207-235.

[28] Shaaban, E., Helmy, Y., Khedr, A., Nasr, M. (2012). A proposed churn prediction model. International Journal of Engineering Research and Applications, 2(4), 693-697.

[29] Vafeiadis, T., Diamantaras, K. I., Sarigiannidis, G., Chatzisavvas, K. C. (2015). A comparison of machine learning techniques for customer churn prediction. Simulation Modelling Practice and Theory, 55, 1-9

[30] Yabas, U., Cankaya, H. C., Ince, T. (2012, July). Customer churn prediction for telecom services. In 2012 IEEE 36th Annual Computer Software and Applications Conference (pp. 358-359). IEEE.